



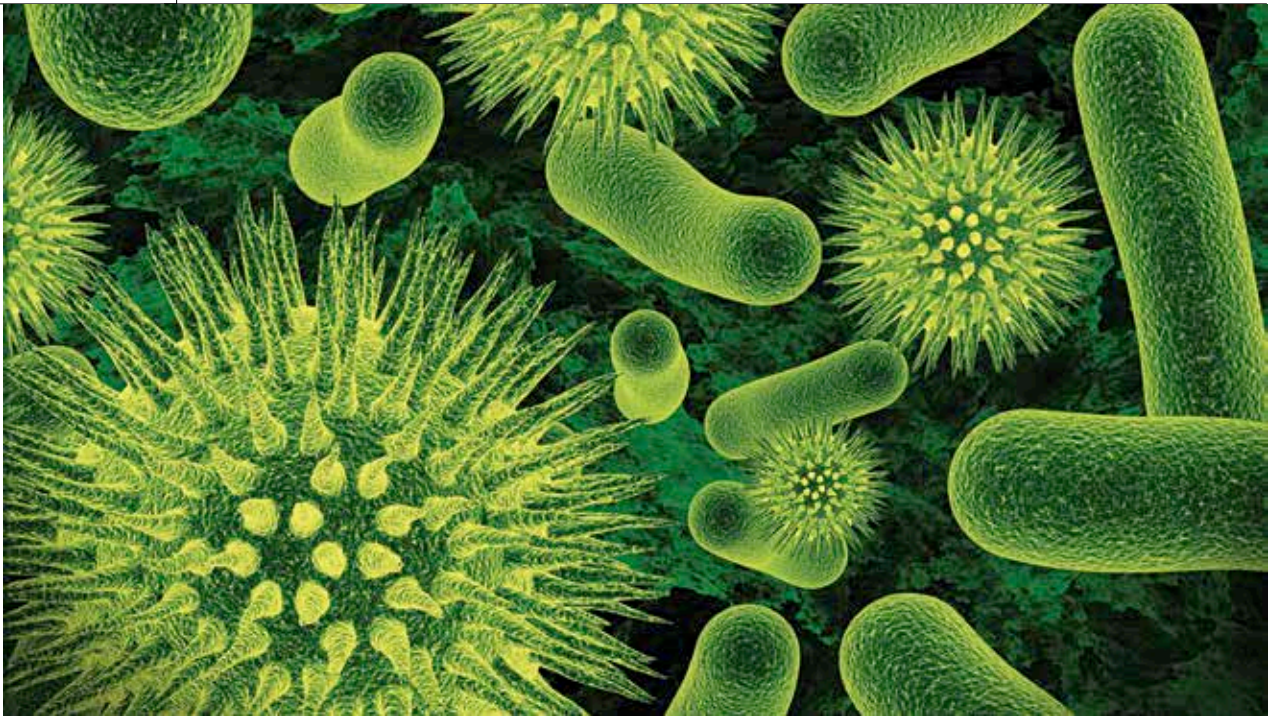
Capuchin monkeys' tool use

Capuchin monkeys have been using carefully chosen stones as tools for the last 3,000 years. <https://digit.in/july19-49>



Puppy Dog Eyes

A research study explains how humans might be the reason dogs evolved to learn the "puppy dog eyes". <https://digit.in/july19-50>



MAN VS PATHOGENS: THE ARMS RACE

The microscopic battle between pathogens and human immune systems is dynamic

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he intimate relationship between pathogens and human hosts has been long-investigat-

ed and is often reckoned as an example of a co-evolutionary arm-race. The problem is that humans tend not to follow the saying 'prevention is better than cure' and only react to extreme crisis. Infectious diseases reportedly claim the lives of 14 million people each year. Reactive human nature tends to undercut the best shot we have at combating these diseases - scientific research.

HISTORY

Infectious diseases have had a colossal impact on our history. They have led to the subsequent rise and fall of civilizations. More military campaigns have been lost or won due to infectious diseases rather than the strategic astuteness of the armies' commanders. In the History of the Peloponnesian War, written by Thucydides in the 5th century BC, it was reported that the plague of Athens devastated the city-state during the second year of the Peloponnesian War in 430 BC when it was at the cusp of victory against Sparta. The fall of the Roman Empire was also said to be due to another epidemic. Infectious diseases caused by disease-carrying pathogens have also played an integral role in past human migrations. In the 16th century, the conquistadores overthrew large swaths of the Americas. This was majorly aided by the diseases they brought along with them like measles and smallpox. Indigenous populations

in these areas had limited immunity to these diseases.

THE PRESENT SITUATION

Fortunately, in the developed world, infectious diseases are no longer one of the leading causes of mortality. They are replaced by age-associated disorders such as cancer, heart disease, and diabetes. Diseases that once had devastating effects on the human population, such as smallpox, have now been exterminated. Other infectious diseases like plague or leprosy that were devastating before, are now mostly controlled, excepting a few hotspots.

Despite these positive advances, the current situation presents new challenges. Globalisation and increased mobility, especially air travel, have further exaggerated the transmission of infectious diseases. The recent outbreak of Zika in North and South America can be ascribed in part to increase of air travel, which brought in passengers from infected areas into Brazilian airports. Both,